SkillBridge (PBL) — Installation Manual

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Document Purpose
This manual explains how to install, configure, and run the SkillBridge project locally and on a server.

Intended Audience: Developers / Installers / Team members responsible for deployment.
Version: 1.0
Date: {Insert date}

1. Overview

SkillBridge is a web application used to support student profiles, job role matching, and placement-related dashboards. It typically consists of:

- Frontend (UI): React / Web UI
- Backend (API): Node.js / Express (or similar)
- Database: MongoDB (common for full-stack projects) or another supported DB

Note: If your repository uses a different stack (e.g., Django/Flask, PHP, PostgreSQL), replace the backend/database sections accordingly.

2. System Requirements

- CPU: Dual-core
- RAM: 8 GB (16 GB recommended)
- Storage: 20 GB free space
- Node.js: LTS version (recommended)
- npm or yarn: latest stable
- Database: MongoDB (if used)
- Browser: Chrome/Edge (latest recommended)

3. Prerequisites & Access

- Access to the project source code repository (Git)
- MongoDB connection details (local or cloud) — if the backend uses MongoDB

- An API key / credentials for any external services (if used) such as: email service, LLM provider, file storage, maps, etc.
- A domain and SSL certificate if deploying to production (optional but recommended)

4. Repository Structure (Typical)

A common structure for full-stack projects is:

```
root/
  client/         (frontend)
  server/         (backend)
  .env            (optional root env)
  README.md
```

If your structure differs, follow the environment file and package scripts in your repository.

5. Installation (Local Development)

5.1 Clone the Repository

```
git clone <REPOSITORY_URL>
cd <REPOSITORY_NAME>
```

5.2 Install Backend Dependencies

```
cd server
npm install
```

5.3 Create Backend Environment File (.env)

In the server folder, create a file named .env.

Copy values from .env.example if available; otherwise use this template (edit keys to match your code):

```
# Server
PORT=5000
NODE_ENV=development

# Database
MONGO_URI=mongodb://localhost:27017/skillbridge

# Auth (example)
JWT_SECRET=<your_secret>
JWT_EXPIRES_IN=7d

# CORS (example)
CLIENT_ORIGIN=http://localhost:5173
```

5.4 Start Backend

```
npm run dev

(or)
npm start
```

5.5 Install Frontend Dependencies

```
cd ../client
npm install
```

5.6 Create Frontend Environment File (.env)

In the client folder, create .env (or .env.development depending on your tooling).

Edit values to match your backend URL/port (example below):

```
# Frontend
VITE_API_URL=http://localhost:5000

# If using Create React App (CRA) use REACT_APP_ prefix
# REACT_APP_API_URL=http://localhost:5000
```

5.7 Start Frontend

```
npm run dev

(or)
npm start
```

Open your browser:
- Frontend: http://localhost:5173 (or as shown in terminal)
- Backend health (if available): http://localhost:5000/ (or /health)

6. Database Setup

6.1 MongoDB (If Used)

- Install MongoDB Community Server on your system and start the MongoDB service.
- Or use MongoDB Atlas (cloud). Replace MONGO_URI in server/.env accordingly.
- Ensure the database name in MONGO_URI exists or allow backend to create it on first run.

6.2 Verify Database Connection

Start backend and confirm logs show a successful DB connection.

If you see errors like "ECONNREFUSED" or "authentication failed", verify:
- MONGO_URI value
- MongoDB service status / firewall rules
- Atlas username/password and IP whitelist (if using Atlas)

7. Running the Application

For local usage, run both backend and frontend in separate terminals.

Recommended order:
1) Start MongoDB (if local)
2) Start backend (server)
3) Start frontend (client)

Login / Use Cases (example based on your screenshots):
- Student Profile page
- Job role matching/dashboard
- Job requisitions list

Note: Exact routes depend on your frontend implementation.

8. Production Deployment (Server Setup)

This section describes a common deployment approach. Your project may use Docker, PM2, Nginx, or a platform like Render/Vercel/Netlify. Adapt as needed.

8.1 Build Frontend

```
cd client
npm run build
```

8.2 Run Backend with Process Manager (PM2 Example)

```
# Install PM2 globally
npm install -g pm2

cd server
pm2 start npm --name "skillbridge-backend" -- start
```

- Set NODE_ENV=production on server.
- Update CLIENT_ORIGIN in server/.env to your production frontend domain.
- Use a reverse proxy (Nginx) for HTTPS and to route requests to backend.
- Do NOT commit .env files. Store them securely in your deployment environment.

9. Troubleshooting

Issue	Possible Cause	Solution
Backend fails to start	Missing/incorrect environment variables	Check server/.env (PORT, MONGO_URI, JWT_SECRET, etc.). Ensure values match your code and are present.
CORS error in browser	client origin not allowed in backend	Set CLIENT_ORIGIN (or CORS config) to the frontend URL/port used by your environment.

Database connection error	MongoDB not running or URI wrong	Start MongoDB service (local) or verify MongoDB Atlas connection string, username/password, IP whitelist.
Frontend shows blank page	API URL incorrect or backend not reachable	Verify VITE_API_URL / REACT_APP_API_URL matches backend host and port; confirm backend is running.
npm run dev crashes	Node version mismatch or missing dependency	Update to an LTS Node.js version; delete node_modules and reinstall: rm -rf node_modules && npm install.

10. Security & Maintenance

- Never store secrets (JWT_SECRET, API keys) in the repository.
- Restrict MongoDB access (local firewall or Atlas IP allowlist).
- Use HTTPS in production.
- Keep dependencies updated and run tests before releases.
- Create backups of your database if used in production.

11. Appendix

A. Environment Variables Checklist
Before starting the application ensure these exist:
• server/.env: PORT, MONGO_URI, JWT_SECRET (and others used by your code)
• client/.env: API URL (and any other keys used by frontend)

B. Support
If installation fails, collect:
• Backend logs (console output)
• Frontend browser console errors
• The exact .env values used (remove secrets if sharing publicly)